



Fig. 9: Laser-diode drive circuit for constant-output power

constituted by R_3 and R_4 and also assuming that $R_5 = R_6 = R_7$. This is true only if voltage V after potential divider arrangement R_{13} and R_{14} and appearing at output of op-amp A_3 did not interfere. This is true as long as diode D_2 is reverse biased, which it is during normal operation.

If the drive current exceeds a certain preset limit governed by voltage at the output of op-amp A_7 , diode D_2 gets forward biased and clamps the voltage at the noninverting input of op-amp A_4 to the voltage at the output of op-amp A_7 , thereby providing overcurrent limit. The circuit may be modified to include a JFET and associated feedback loop components as shown earlier in Fig. 6 to improve current stability.

Laser-diode drive circuit: constant-output-power mode

Laser diode can be driven to maintain a constant-output power by varying the drive current in accordance with the output power. The feedback loop circuit is such that it increases or decreases the drive current in response to decrease or increase in output power. Fig. 9 shows the constant-output-power drive circuit. The integral photodiode produces current proportional to optical output power from the laser diode. This photocurrent is converted

into a proportional voltage using a transimpedance amplifier configured around op-amp A_4 .

The voltage representative of laser power is then summed up with a reference voltage representing the desired power level in an inverting summer configuration A_5 . The output of the summer, which is null when the actual power level equals the desired power level, feeds integrator A_6 that provides the correction signal even for infinitesimally small deviations from the desired power level. The integrator output summed up with a bias voltage feeds control element, which is a junction FET in this case.

The drive current, and hence the output power, is governed by voltage present at noninverting input of op-amp A_3 . Evidently, reduction in laser-diode-output power causes reduction in photodiode current, which finally leads to gate terminal of JFET becoming more negative. This further causes a reduction in voltage at A_3 noninverting input, increasing the drive current to restore the output power. Similarly, increase in laser-diode power reduces drive current to restore the power at nominal value.

Driving laser diode in pulsed mode

There are two possible modes of operation of laser diodes to produce a

pulsed output. In one of the operational modes of relevance to their use in military devices, such as laser rangefinders employing time-of-flight principle and laser proximity sensors, the laser diode is driven by current pulses that are a few tens to a few hundreds of nano-seconds wide. In the other operational mode, called quasi-CW mode, laser diode is driven by current pulses that are typically hundreds of micro-seconds to a few milliseconds wide. This operational mode is invariably used in

the case of laser-diode arrays pumping solid-state lasers, including those for military rangefinders, target designators, electro-optic countermeasures and so on.

Quasi-CW operation of laser diodes for optical pumping of solid-state lasers is at relatively low repetition rates of typically a few hertz to a few tens of hertz. This allows them to be operated at relatively high peak powers, which is made possible due to low duty cycle of operation of quasi-CW devices, which keeps the average power low.

An important consideration while designing laser-diode drive circuits for pulsed operation, conventional or quasi-CW operation, the laser diode(s) must not be switched between cut-off and the nominal maximum value. This is highly detrimental to the life of laser diodes. They must always be operated between lower values slightly greater than the lasing threshold and the nominal value. Laser diode driver circuits for conventional pulsed and quasi-CW operation are configured around the same basic building blocks as those described earlier in the case of CW operation. The laser diode drive circuit of Fig. 8 can be used for operation in conventional pulsed mode by applying the desired pulsed waveform at the modulation input. The lower value of drive current is governed by